

Masimba Collins

Automation and Robotics Engineer



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in [CollinsMasimba](#)

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Skills

- **Programming:** Python, Pytest, C++, ROS2, MATLAB/Simulink, OpenCV, Github, FoE (File over EtherCat), Siemens TIA Portal, TwinCAT, SCADA, DCS, HMI Design and Visual Components, Linux, Docker/Podman, RTOS.
- **Robotics frameworks & Tools:** ROS2, Ubuntu, Gazebo, RViz, Mujoco.
- **Sensors and Hardware:** LiDAR, Radar, Proximity Sensors, KUKA Robots, ARM Cortex Microcontrollers, Actuators (Servo Motors, Stepper Motors), MES interfaces.
- **Soft Skills:** Strong interpersonal communication, proactive problem-solving, self-motivated team player, Customer & Application Support.

Research & Publication

- Water Treatment and Plant Automation.
- Autonomous Robot Motion Planning.
- Kuka Manipulator Adaptive Control
- Conveyor Belt Automation

Portfolio

- [Github Portfolio](#)

Zertifikate

- **Microcontroller Certification:** University California, Irvine.

Activities

- **KBTU Engineering Club:** Participated in Eco-Shell 2024.

Languages

- **English** C2.
- **German** B2

Professional Summary

Automation and Robotics Engineer with 2+ years focused on microcontroller-based motion control, servo drive firmware and PLC Programming. At Synapticon in Stuttgart, developed and debugged PID/FOC algorithms for motion control firmware, resolved complex EtherCAT/PROFINET integration issues in Siemens TIA Portal, implemented secure FoE firmware updates, and optimized torque/position feedback loops via HIL simulations. Developed ROS2 packages for Safe Motion Module (SMM) drives in testing and validation. In research at Kazakh-British Technical University, enhanced KUKA manipulator precision through Deep-Q learning integrated with embedded control systems. Proficient in C/C++, RTOS, and cross-team firmware validation ready to contribute to compact servo controller platform.

Experience

June 2025- Present

Robotics Engineer Internship

Synapticon Gmbh Schönaich, Germany

- Developed python, bash and pytest scripts for automated testing motion control firmware (PID and Field-Oriented Control) using RestAPI framework.
- Developed ROS2 packages in C++ and Python for testing the control of SMM (Safe Motion Module) drives for Positron Humanoid and PMSMs for the integros.
- Reduced integration cycle times by 20% by scaling test coverage and optimimizing test execution in TWINCAT 3 PLC for Synapticon Arm Robots.
- Designed and implemented test suite for Bumpless transfer in robot control motion in TWINCAT using Structure Text (ST) for PMSMs for integros.
- Debugged complex firmware defects in Motor Control Behavior for stability and torque response using JTAG.
- Built and managed Continuous Integration (CI/CD) pipelines using Jenkins, Docker, and GitHub to automate build, test, and deployment phases for firmware testing.
- Investigated and fixed critical build/integration failures involving real-time fieldbus communication (EtherCAT master/slave) and PROFINET integration layers.
- Collaborated with mechatronics teams to integrate HIL simulations (MATLAB/Simulink) and bench testing for actuator validation based on the V-model.
- Improved torque and position feedback validation by correlating HIL models with bench measurements and automating acceptance criteria.

October 2023- June 2025

Robotics Research Assistant

Kazakh-British Technical University| Almaty, Kazakhstan

- Designed Modicon M221 PLC + SCADA system for water treatment plant, cutting energy use 3.5% and achieving 98% uptime via optimized pump logic.
- Developed OpenCV motion planning algorithm in ROS2 for autonomous robot, boosting efficiency 5.56% (from 0.90 to 0.95) and improving motion precision and safety.
- Developed a hierarchical diffusion-based framework for trajectory planning in mobile robots using SLAM in ROS2.
- Implemented Siemens S7-1200 PLC + TIA Portal HMI for conveyor automation, raising throughput 10% and reducing fault detection time by 4 seconds .
- Investigated DQN-EKF fusion for robust state estimation and decision-making in uncertain robotic environments using reinforcement learning and Extended Kalman Filter.
- Applied Deep Q-Learning to KUKA manipulators, improving end-effector precision 20% (from 0.25 mm to 0.20 mm) with adaptive policies integrated into Siemens TIA Portal control.

Education

September 2022- Jan 2026

Bsc Automation and Control Engineering

Kazakh-British Technical University| Almaty, Kazakhstan

Relevant Coursework: Modelling and Control of Robotic Manipulator, Control Theory and Application, Scientific Programming With MATLAB, Computer Systems, Advanced Engineering Mathematics, Control Theory and Application, Mobile Robots, Network Mobile Robot Systems, Process Control Lab, Online Problems, Automated Driving, 3D Computer Vision, Intensive Course in Statistics for Researchers in Engineering Sciences, Application of Robots, Siemens PLC: TIA Portal, Machine Vision and Image Processing, Automation and Intelligent System. Embedded System using Raspberry Pi and Jetson